

THE BONDI DIPOLE IN FULL NUMERICAL RELATIVITY: A SELF-ACCELERATING POSITIVE–NEGATIVE MASS BINARY

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Bondi showed in 1957 that bodies of opposite *active* gravitational mass self-accelerate: the negative chases the positive it repels, and the pair runs off together. We evolve this “Bondi dipole” in 3+1 numerical relativity: two complex scalars share identical Klein–Gordon dynamics; only the phantom enters Einstein’s equations with a minus sign—inertial and passive masses positive, active mass negative. Across a matrix of thirty-four evolutions—runaway pairs, controls, and parameter scans—a mass-matched pair released at rest accelerates as a unit. The midpoint moves 3.00 ± 0.01 by $t = 200$ with the separation held to 1%, and reaches a speed of $0.056c$ by $t = 400$ with the acceleration steady to 2%; the total signed momentum holds at zero to $\lesssim 1\%$. The force is gravity on both of its axes: $a \propto d^{-2.03 \pm 0.01}$ over $d = 8\text{--}20$, $a \propto M^{0.97 \pm 0.06}$ over a factor 2.5 in mass, and $a d^2 / \bar{M} = 1$ within 2.4% on the equal-mass ladder. Swapping the sectors inverts the acceleration to two parts in 10^5 ; gauge, solver-depth and mesh variations move the drift by $\lesssim 0.01\%$ and box doubling by 4%; same-sign control pairs hold their centroids to $\lesssim 8 \times 10^{-4}$ even while merging. The runaway carries no detectable gravitational radiation: the signed dipole cannot radiate, the quadrupole’s $\dot{Q}$ is constant, and the measured $\ell = 2$ amplitude falls as $r^{-4.8}$ —near zone, not flux. The phantom star is, to our knowledge, the first asymptotically flat body of negative ADM mass evolved in numerical relativity; alone it survives to $t = 1000$, slowly relaxing outward.

I. INTRODUCTION

Bondi [1] gave negative mass its modern treatment, distinguishing inertial, passive and active gravitational mass and noting the consequence of mixing signs: a negative-mass body is *attracted* by a positive one, the positive body is *repelled* by the negative one, and the pair runs off together (Fig. 1)—momentum conserved, because the negative body’s contribution is negative. Bonnor and Swaminarayan [2] found exact solutions for uniformly accelerated opposite-mass pairs, later generalized in Ref. [3]; Forward [5] showed “in tedious detail” that the self-accelerating pair violates neither momentum nor energy conservation, for equal and unequal mass magnitudes alike, so objections to negative mass must be sought elsewhere [6, 7]. Newtonian N -body work has since carried the same sign rules into cosmological structure formation [42, 43].

Why put this in full numerical relativity? Because nothing in Einstein’s equations forces mass to be positive—positivity is a theorem only once energy conditions are imposed [27, 28]—and modern physics keeps producing sources that strain those conditions: Casimir energies, quantum fields in curved spacetime, phantom dark energy [8]. Every traversable wormhole [9] and warp spacetime [10] needs matter of exactly this kind—the ghost scalar holding a wormhole throat open violates the same energy condition, and gravitates with the same sign, as the phantom sector evolved here, though with inverted field dynamics (Sec. II)—and the standing objection to all of them is Bondi’s runaway itself—matter that accelerates for free looks like a perpetual-motion machine [6, 7].

each body is accelerated by the *other’s* active mass; inertia is $|M|$ and positive in every case

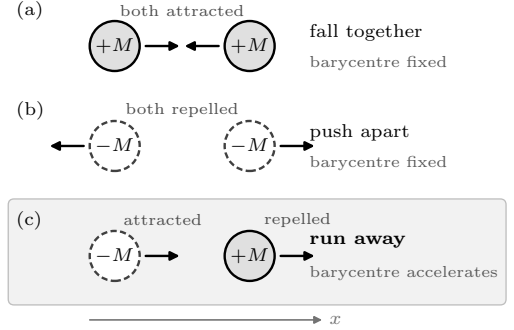


FIG. 1. Bondi’s sign rules for two bodies released at rest, in the style of Forward’s analysis [5]. Every body has positive inertial and passive mass; only the *active* (source) mass carries a sign—negative for the dashed bodies. (a) Two positive-mass stars attract and fall together; (b) two negative-mass stars repel; in both the forces oppose and the pair barycentre stays put. (c) The mixed pair—the Bondi dipole: both forces point the *same* way, so the pair accelerates while the system carries zero total momentum in the signed bookkeeping. Each row is a run family of Table I; in our field-theoretic realization the same-sign rows additionally feel their shared field’s own short-range attraction (Sec. VE), which the mixed row, by construction, cannot.

That objection has only ever been examined for point particles and prescribed metrics; the nonlinear evolutions that exist for energy-condition-violating matter—ghost-scalar wormholes in spherical symmetry [15, 30], static phantom-field stars [31], and the 3+1 collapse of an Alcubierre warp bubble [11]—concern single objects. Whether the runaway survives full nonlinear gravity—finite bod-

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ies, backreaction, radiation—is a checkable question, and we are not aware of a previous 3+1 evolution of a gravitationally bound positive–negative active-mass pair with dynamical, constraint-solved matter.

The question has acquired observational stakes: negative-mass binaries have been proposed as sources of anomalous gravitational-wave signatures—anti-chirps, dispersal, runaway motion—turning Bondi’s paradox into an exclusion channel for real detectors [39], while Nojiri and Odintsov [13] find that a bound positive–negative system *can* form for a compact mass immersed in a cosmological fluid with negative cosmological constant. Our asymptotically flat, isolated pair is the complementary case, and it does not bind—it runs away.

The programme: a minimal field-theoretic realization of Bondi’s sign structure (Sec. II); dressed soliton stars of both signs as constraint-satisfying initial data, with the two construction requirements that decide whether the measurement is possible at all (Sec. III); a falsifiable matrix of thirty-four evolutions, each a *cell* of that matrix (Sec. IV); the measurement—a constant-rate, zero-momentum, constant-separation runaway obeying $a = \dot{M}/d^2$, with every control null (Sec. V); the gravitational-wave bound on a doubled box (Sec. VI); and the systematics with their measured sizes (Sec. VII). Every evolution ran on GPUs with `GRTEclyn` [24], the `AMReX`-based [26] GPU port of `GRChombo` [23]; with the wormhole campaign of Ref. [45] these are, to our knowledge, the first evolutions of energy-condition-violating matter in GPU-native numerical relativity, the warp-bubble collapse of Ref. [11] being the closest CPU-side precedent. Throughout, $G = c = 1$, the scalar mass sets the unit ($m = 1$), X is the pair midpoint (the mean of the two sector barycentres), $\Delta X = X(t) - X(0)$ its drift, and d the instantaneous separation between the two stars—the *gap*. Runs are named for the sectors they contain, P for a canonical star and M for a phantom one: PM is the mixed pair, PP and MM the same-sign controls, MP the mixed pair with the sectors swapped in place.

II. THE BICOMPLEX SCALAR MODEL

The matter must keep canonical *dynamics*—positive inertial and passive mass—while only its gravitational *source* changes sign; it must form stable, localized stars; and the negative-mass star must be held together by something other than gravity, since its own gravity pushes it apart. Each obvious candidate fails one of these: the negative-mass Schwarzschild spacetime is a naked singularity with exponentially growing smooth linear perturbations [14] and contains no matter from which to build initial data; a negative-energy fluid has no equilibrium to sit in, Bonnor’s static spheres [4] demanding an inverted interior with no non-gravitational restoring force; and a *kinetic-term* ghost—wrong-sign kinetic term, potential untouched—carries its sign flip into the Klein–Gordon equation, where it inverts the effective potential

and unbinds the star, and flips inertial and passive mass along with the active one (Forward’s all-negative matter [5], not the signature under test); such fields collapse or explode in evolution [15, 30, 45]. The phantom used here is the other kind of sign flip: an *overall*-sign sector, defined in Eq. (5), whose field equations are untouched and whose stress tensor alone is negated.

What passes is a *soliton*: matter self-bound by its own interactions, for which gravity is a dressing rather than the binding agent. We take two independent complex scalars, Φ_+ (*canonical*) and Φ_- (*phantom*), on one spacetime—a *bicomplex* field in the sense used throughout this paper, meaning two independent complex sectors and not a bicomplex-number-valued field—sharing the standard sextic potential of Q-ball and solitonic boson-star studies [16–18],

$$V(|\Phi|^2) = \frac{1}{2}m^2|\Phi|^2 - \frac{1}{4}\lambda|\Phi|^4 + \frac{1}{6}\mu|\Phi|^6, \quad (1)$$

with $\lambda, \mu > 0$: the mass term makes the field oscillate, the attractive quartic lets it clump, the repulsive sextic stops the clumping short of collapse. Sharing one potential makes the matter physics of the two sectors *identical*, so the single sign introduced next is the only possible cause of any asymmetry the simulations show.

That sign lives in one place—the Einstein equations are sourced by the *signed* sum of two canonical stress tensors,

$$G_{\mu\nu} = 8\pi (T_{\mu\nu}[\Phi_+] - T_{\mu\nu}[\Phi_-]), \quad (2)$$

while the field equations of *both* sectors keep their canonical form,

$$\square \Phi_{\pm} = 2 \frac{\partial V}{\partial |\Phi|^2} \Phi_{\pm}, \quad (3)$$

the factor 2 fixed by the $\frac{1}{2}$ -normalized kinetic term. In 3+1 language the sources seen by normal observers accumulate per sector with a sign $\epsilon_{\pm} = \pm 1$,

$$\rho = \sum_{s=\pm} \epsilon_s \rho[\Phi_s], \quad S_i = \sum_{s=\pm} \epsilon_s S_i[\Phi_s], \quad (4)$$

and likewise for S_{ij} , each bracket being the standard scalar expression. The model derives from

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi} + \mathcal{L}[\Phi_+] - \mathcal{L}[\Phi_-] \right], \quad (5)$$

with $\mathcal{L}[\Phi] = -\frac{1}{2}\nabla_{\mu}\Phi^*\nabla^{\mu}\Phi - V(|\Phi|^2)$: a sector’s overall Lagrangian sign cancels from its Euler–Lagrange equations—hence Eq. (3)—but survives the metric variation—hence Eq. (2). Bianchi consistency is automatic (each sector’s stress tensor is separately conserved on shell, so the signed sum is too); well-posedness is inherited (the sector sign enters CCZ4 only through algebraic source terms); and the mass signature is Bondi’s: the phantom *moves* canonically—positive inertial and passive mass—but *gravitates* with the minus sign of Eq. (2), so its active mass, and its ADM mass, are negative.

The model has exactly two interaction channels, and the run matrix is built to isolate them. The action carries no cross term between Φ_+ and Φ_- —each potential depends on its own $|\Phi_s|^2$ alone—so **two stars of different sectors interact only through the metric**: gravity is the mixed pair’s only channel, by construction. Two stars of the *same* sector, by contrast, live in one complex field, whose self-interaction gives overlapping lumps a direct, short-range attraction on top of gravity. Section V E measures that field channel—it is strong, and blind to the sign of the mass—and it is precisely why the mixed pair, which cannot have it, is the clean Bondi configuration.

The phantom sector violates the null energy condition by construction, and quantum viability is not claimed—a sector entering gravity with negative energy admits graviton-mediated vacuum decay [32, 33]—so the model is a classical probe of *if such matter existed, what would gravity do*, in the spirit of phantom cosmology [8].

III. INITIAL DATA: DRESSED SOLITONS

A. The stars

A soliton of this theory oscillates coherently, $\Phi \propto e^{-i\omega t}$, and ω labels the solution family: $\omega = m$ is the free-field limit, $1 - \omega$ the binding energy per unit mass. The ratio λ^2/μ sets the lowest attainable frequency, $\omega_{\min} = \sqrt{1 - 3\lambda^2/16\mu}$ [17]; all runs use $\lambda^2/\mu = 4.8$ ($\lambda = 10240$, $\mu \simeq 2.18 \times 10^7$, amplitude $\simeq 0.019$), so $\omega_{\min} = 0.316$.

A flat-space soliton cannot simply be placed on an initial slice: the constraints are violated by a bare Q-ball, and once its own gravity acts the profile is no longer an eigenstate. The star we need is solved with its gravity *on*: the boson-star problem [19–21] with a sextic self-interaction, posed for both source signs. With $\Phi = \varphi_0(r)e^{-i\omega t}$ and a conformally flat, maximally sliced metric, Eqs. (2)–(3) reduce to coupled ODEs for $(\psi, \alpha, \varphi_0)$; **the phantom star is obtained by flipping the sign of the matter sources in the metric equations only**, the Klein–Gordon line keeping its canonical form. The result is a self-interaction-bound soliton with repulsive gravity: $\psi < 1$, $\alpha > 1$ in the core, and negative ADM mass. We shoot at fixed frequency, bisecting the central amplitude and outer-iterating until ω/α_∞ matches the request to $\sim 10^{-5}$.

Figure 2 shows both families. The campaign pair takes the canonical star at $\omega = 0.750$, $M_{\text{ADM}} = +0.014350$, and the phantom at $\omega = 0.7603$, $M_{\text{ADM}} = -0.014295$: mass-matched to 0.39%, because Bondi’s point-mass limit is governed by each body’s exterior field, hence its ADM mass, alone. The pair mean $\bar{M} \equiv \frac{1}{2}(M_+ + |M_-|) = 0.014322$ is what the midpoint acceleration measures. The canonical star has rms radius 4.34 and compactness $M/R \sim 3 \times 10^{-3}$ —two orders below what any horizon needs—and a four-point survey ($\omega = 0.75\text{--}0.90$, $t = 120$) confirms the family static where it is used: the minimum

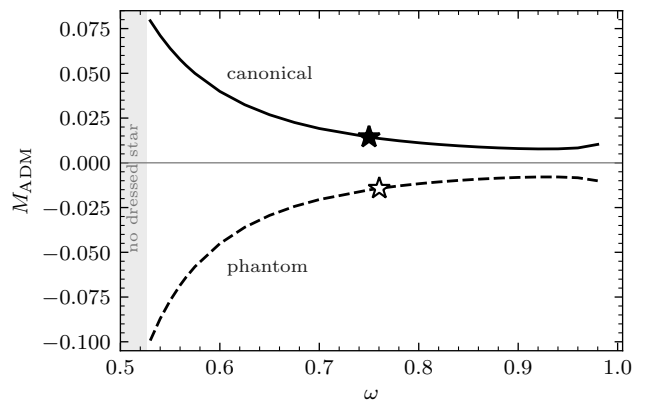


FIG. 2. ADM masses of the spherical stars against oscillation frequency at the working couplings. Lower ω means deeper binding and larger $|M_{\text{ADM}}|$; at equal frequency the phantom star is the heavier in magnitude, its repulsive self-gravity spreading it over more volume. Stars: the campaign pair, mass-matched to 0.4%. Both branches floor at $\simeq 0.54$ of the campaign mass near $\omega \simeq 0.93$, and neither has a bound star at $\omega = 1$ —the bound that shapes the mass ladder of Sec. V F.

lapse moves by at most 1.2×10^{-3} from birth. Each star is seeded from its own profile table with its own frequency, and the conjugate momentum carries the solved lapse, $\Pi = -(\omega/\alpha(r))\varphi_0(r)$, so the harmonic phase starts on its eigenstate.

B. Two construction requirements

GRTresna [25] then re-solves the constraints for the pair (CTTK family [35], sources summed with their signs star by star). Two choices in this step decide whether the runaway is measurable at all. Both were established by direct testing.

Both sectors must be built on the same slicing. The CTTK algebraic ansatz ties the initial K to the local energy density, $K \propto \sqrt{\rho}$ —imaginary for $\rho < 0$, so a phantom sector forces the maximal-slicing ($K = 0$) path. Letting the canonical sector default to the algebraic path builds the two stars by *different methods*: the canonical star is then born inside a coherent contraction ($\max|K| \approx 0.1$ at $t = 0$, four orders of magnitude above the phantom’s) and squeezed through its first two field periods. We therefore construct *both* sectors with $K = 0$, which for this matter also converges better; the birth kick vanishes ($\max|K| \rightarrow 10^{-5}$, landing on the phantom’s value) and the two constructions differ in nothing but the sign of ρ .

The solve grid must share the evolution grid’s grid-cell centres. The elliptic solve runs on its own (wider) box; if its spacing differs from the evolution’s, the solved metric lands on the evolution grid displaced by a fraction of a grid cell while the matter is repainted exactly—so every star is born off the centre of its own well, the canonical

falling back down it and the phantom pushed up it, a directional artefact that does not converge away and was comparable to the signal. Matching the spacings makes the transfer a straight copy: the solved metric moves grid cell by grid cell (a piecewise-constant transfer, exact once the grid-cell centres coincide) while the matter is repainted analytically from its radial profile table. The metric-minus-matter centroid offset is then zero to the diagnostic’s 10^{-4} precision on every cell here, and the lone-star drift below is $269\times$ smaller than on mismatched grids. All solves run at the evolution spacing on a doubled box ($L = 128$); exit residuals, tightened with the grid, are in Appendix A.

Every pair places its lumps at $x = \pm d_0/2$ and paints both from their radial tables with the same internal $U(1)$ phase: all pairs are seeded *in phase*, relative phase $\delta = 0$ (the packed launch parameter `phase0` is the placement azimuth, 0 and π , not the field phase). For the mixed pair δ is pure gauge, the two sectors being independent fields with no cross term. For the same-sector controls it is not: two Q-balls of one field attract through the shared field at $\delta = 0$ and repel at $\delta = \pi$ [40, 41], so the seeding used here is the attracting configuration, consistent with the merger those cells show (Sec. V E). Superposed same-field data also carries known phase-dependent constraint systematics that the re-solve reduces but does not eliminate [34]—one reason those cells serve as momentum nulls and not as precision force measurements.

IV. THE CAMPAIGN

A. Numerics

Evolutions use GRTeclyn: CCZ4 [22] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, $1 + \log$ slicing [36], a Gamma-driver shift [37, 38] with $\eta = 1$, fourth-order stencils, RK4, Kreiss–Oliger dissipation $\sigma = 2$, Sommerfeld outer boundaries, and—because massive-field radiation reflects from massless-wave boundaries—an absorbing sponge layer over $r = 24$ –32 (moved to 48–60 on the doubled box). The sponge works: the net inward flux through the accounting boundary stays at round-off ($\lesssim 10^{-7}$) even in the violent same-sign cells. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor and the dominant cost of the campaign (Appendix B); it is not conservatism but the sextic self-interaction’s own stability limit—at $\Delta t = 0.2 \Delta x$ the star disperses outright.

Every cell is *strictly uniform-grid* (`max_level` = 0), so the resolution ladder means exactly one thing—the grid spacing—and the aligned-grid handoff of Sec. III B is preserved verbatim. The main box is $L = 64$ at $N = 128$, 192, 256 ($\Delta x = 0.50, 0.33, 0.25$ —about 9, 13 and 17 points across a stellar rms radius of 4.3); the wave-zone box doubles L at fixed $\Delta x = 0.25$ with extraction shells at $R = 16$ –40. The reference separation is $d_0 = 10$, about twice the 90%-weight radius, and the separation

scan spans $d_0 = 8$ –20. Stars are released *at rest*—no boosts, no driving—so any drift is dynamical. All cells run to $t = 200$ (the stability survey to 120; the long pair run to 400, the lone-phantom control to 1000, and the two late-time follow-up cells of Sec. VII to 784 and 600), and every fitted acceleration a is twice the quadratic coefficient of $X(t)$ over $5 \leq t \leq 200$, the first five units excluded while the gauge settles (Sec. VII for the convention’s percent-level sensitivity).

B. Diagnostics and the run matrix

Each sector is tracked two ways, continuously: a whole-domain barycentre with weight $w_s = (|\Phi_s|^2 + |\Pi_s|^2)^{1/2}$, and a sub-cell core tracker at the sector’s field peak. The headline quantity is the *pair midpoint* $X = \frac{1}{2}(\bar{x}^{(+)} + \bar{x}^{(-)})$: it is what Bondi’s argument predicts to accelerate, it is defined identically for pairs and controls, and by symmetry it is exactly zero for every control—so the control rows of Table I measure the noise floor rather than assume one. The two trackers agree on the pair drift to 1%. Volume-integrated sector momenta, constraint norms, confinement, the $\ell = 2$ modes of Ψ_4 , and energy-condition monitors are recorded alongside. The last return what the model prescribes and nothing more: the minimum NEC is -2.20×10^{-4} in every cell containing a phantom—the lone phantom’s own value, flat over the run—while the lone canonical star stays positive everywhere, its minimum $+7 \times 10^{-42}$ —zero to round-off in the vacuum tail, no violation at all.

Each row of Table I is a falsifiable prediction of the sign structure.

V. RESULTS

A. The runaway: constant rate, zero momentum, constant separation

Released at rest with the canonical star at $x = 37$ and the phantom at $x = 27$, **both stars accelerate toward $+x$ as a unit** (Figs. 3 and 4). On the finest grid the midpoint moves $\Delta X = +3.0016$ by $t = 200$ at fitted acceleration $a = 1.549 \times 10^{-4}$; a single parabola fits the whole trajectory. The $t = 400$ run (base grid, where $a = 1.45 \times 10^{-4}$) shows the rate is *sustained*: fitted over [133, 266], [300, 400] and [350, 400] the acceleration is 1.451, 1.422 and 1.417×10^{-4} —steady to 2% across a factor three in elapsed time, with the pair at $0.056c$ and still gaining speed linearly when the run ends. There is no impulse, no saturation, and no secondary acceleration.

The two Bondi signatures frame that motion. **Displacement without momentum** (Fig. 4d): each sector’s volume-integrated momentum grows steadily—both reach 5.3×10^{-4} by $t = 200$, the scale set by $M|v|$ —while the *signed* total of Eq. (4) cancels to 3.8×10^{-6} , under 1% of either term, and that residual *converges*

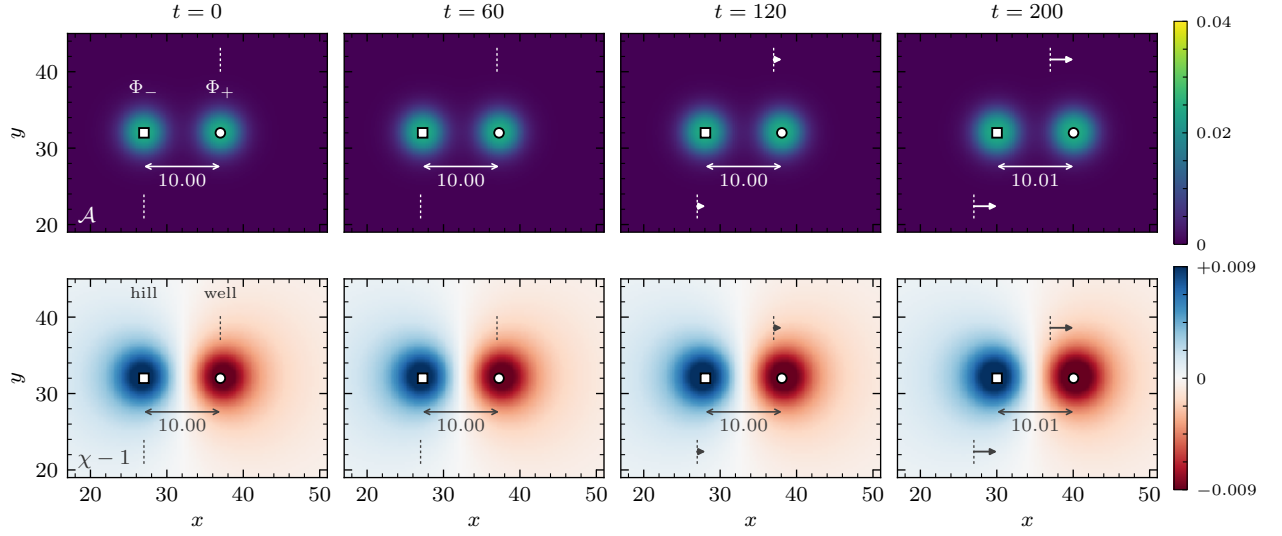


FIG. 3. The runaway in the headline cell (PM, $d_0 = 10$, $\Delta x = 0.25$), $z = 0$ plane, at $t = 0, 60, 120, 200$. Top: scalar-field activity $\mathcal{A} = \sum_s (|\Phi_s|^2 + |\Pi_s|^2)^{1/2}$, both sectors, projected. Bottom: $\chi - 1$; the phantom star is a *hill* ($\chi > 1$, blue, left) and the canonical star a *well* ($\chi < 1$, red, right)—the model’s one sign made visible. Colour bars are locked across stills. Open markers: the tracked cores (circle Φ_+ , square Φ_-); dashed ticks: release positions; arrows: displacement since release; bracketed number: the measured gap. Both stars move toward $+x$ at constant acceleration while the gap holds to 1%. Movie: Video 1; youtu.be/egzPDbz2_B0.

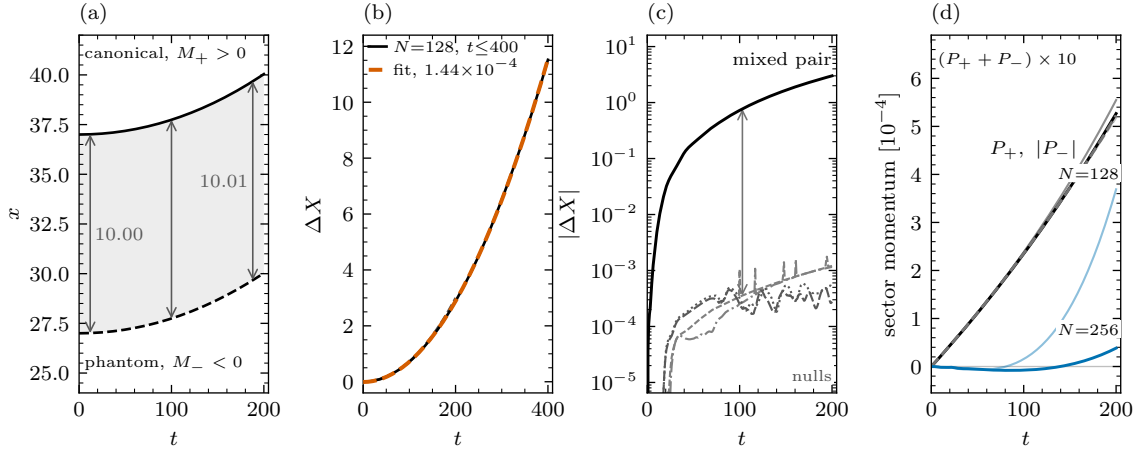


FIG. 4. (a) Core worldlines in the headline cell ($\Delta x = 0.25$): the phantom chases the canonical star it repels; the gap is marked at three times. (b) Midpoint drift of the $t = 400$ run (solid) with the single constant-acceleration fit over $5 \leq t \leq 400$ (orange dashed)—one parabola holds for the whole run—and the finest-grid $t \leq 200$ cell (dashed). (c) The signal against every null on a log scale: the mixed pair’s midpoint displacement over the lone stars’ core drift and the same-sign pairs’ *centroid* displacement (those stars merge; their centroids stay below 8×10^{-4} , Fig. 7b)—225 $\times$ above the like-for-like barycentre floor, $\sim 2000\times$ above the cores (Sec. VE). (d) The Bondi signature itself: each sector’s volume-integrated matter momentum climbs to $\sim 5 \times 10^{-4}$ while the *signed* sum of Eq. (4) (blue, magnified ten times) holds at zero—and holds better on the finer grid, the base rung being the fainter curve. Displacement without momentum, measured. $t = 400$ movie: Video 2; youtu.be/zZcr2N12aww.

toward zero with the grid (3.7×10^{-5} at $N = 128$, 3.8×10^{-6} at $N = 256$), so it is a discretization residual and not a floor. (The cancelling quantity is the matter momentum of Eq. (4), not the ADM momentum—a distinction without numerical consequence at compactness $M/R \sim 3 \times 10^{-3}$.) The pair moves three units while carrying no net momentum, exactly as the signed bookkeeping

demands. **Constant separation:** the barycentre gap ends at 9.915 and the core gap at 10.013, and the *proper* separation between the cores, $\int \sqrt{\gamma_{xx}} dx$ along the axis, runs $10.0001 \rightarrow 10.0244$ —so rigidity holds to 1% over $t \leq 200$ whether it is read in coordinates or in proper distance (the long run opens the gap late, and does so in proper distance too; Sec. VII).

TABLE I. The falsifiable run matrix: thirty-four cells. The sign structure predicts a drifting midpoint for the mixed pair only, reversal under swapping the sectors, and a d^{-2} rate; every control row must return a null *centroid drift*—including the same-sign pairs, which merge into one lump and still do not displace their centroid. ΔX at $t = 200$ unless marked. The separation scan is $d_0 = 8, 12, 16, 20$; the mass ladders are the three retuned pairs at $d_0=10$ and the four equal-mass pairs at $d_0=20$; the stability scan covers $\omega = 0.75\text{--}0.90$. Full cell names and per-cell data are in the results pack (Sec. IX).

run	grid(s)	ΔX	verdict
PM, $d_0=10$	256^3	+3.0016	runaway
PM, $d_0=10$	192^3	+3.0139	converged pair
PM, $d_0=10$	128^3	+2.8815	4% low
PM, $d_0=8\text{--}20$	128^3	—	$a \propto d^{-2.03}$
MP mirror	128^3	−2.8815	reversed ^a
PM, deeper solve	128^3	+2.8815	+0.002%
PM, AMR on	128^3	+2.8815	+0.001%
PM, gauge $\eta=2$	128^3	+2.8813	−0.007%
PM, box doubled	256^3	+2.7606	−4.2%
PM, $t \leq 400$	128^3	+11.518	a steady
PM, $t \leq 784^d$	128^3	—	0.12c reached
PP (+, +) ^b	$128^3\text{--}256^3$	$\leq 7.8 \times 10^{-4}$	merges; null
MM (−, −) ^b	$128^3, 192^3$	$\leq 5.3 \times 10^{-4}$	merges; null
P, M lone stars ^c	128^3	$\leq 1.8 \times 10^{-3}$	null
M alone, $t \leq 1000$	128^3	—	−5.4% peak
mass ladders	128^3	—	$a \propto M^{0.97}$
stability scan	128^3	—	static

^a Drift ratio −1.000022, acceleration ratio −1.000010.

^b Both same-sign pairs coalesce into a single lump (Sec. VE); the entry is the largest pair-centroid excursion *through* that merger—the quantity the null tests. MM has two 128^3 cells; the second adds frame output.

^c Largest core displacement on any axis over $t = 200$; the phantom sits off-centre, the sharper test. Against the tracker the pair drift uses, the phantom’s own barycentre moves 1.3×10^{-2} (Sec. VE).

^d Two late-time follow-ups: a recentring box carried to $t = 784$ and a static-box control started off-centre to $t = 600$, which together exclude the diagnostic box and place the opening on the grid; pair rigidity is quoted only to $t \approx 500$ (Sec. VII). The 0.12c is the leading star’s speed; the midpoint reaches 0.081c.

The matter survives the acceleration. The field peak is flat to 0.3% over the run, the sector rms radii grow only from 4.3 to ~ 5.2 , the minimum lapse never leaves 0.991, and $\min \chi = 0.9889$ —nowhere near collapse ($\chi \rightarrow 0$) or dispersal. A lone star under the same numerics holds its peak to 0.5% for $t = 200$.

B. The force law: $a d^2$ returns the stars’ mass

Bondi’s law is quantitative: each star is accelerated by the *other’s* active mass, so $a = \bar{M}/d^2$. Five separations test it (Fig. 5a,b). A power law through the

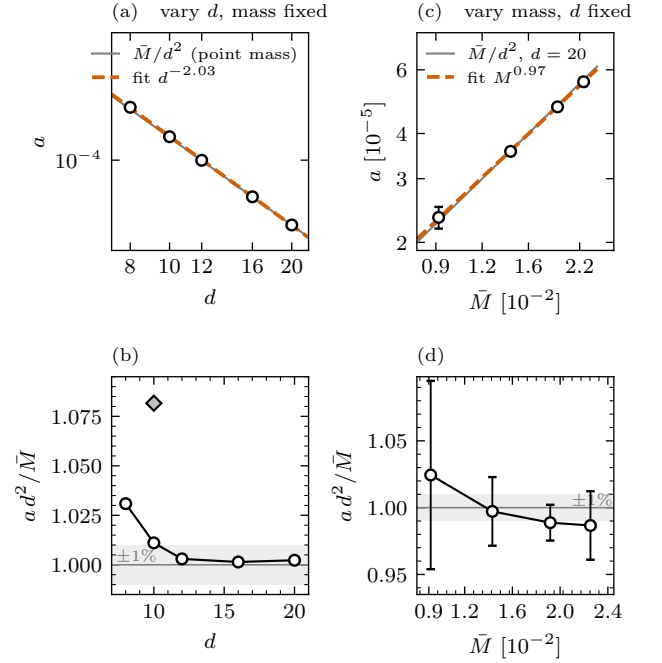


FIG. 5. Both axes of $a = \bar{M}/d^2$, measured. *Left, mass fixed:* (a) fitted acceleration against separation over $d_0 = 8\text{--}20$ at the base grid (open circles); the power-law fit (orange dashed) gives $d^{-2.03 \pm 0.01}$, and the point-mass line $\bar{M}/d^2$ (grey) is indistinguishable from it at this scale. (b) $a d^2$ over the pair mass $\bar{M}$: the close-pair excess decays monotonically, gone by $d = 16$. The open diamond is the converged ($N = 256$) value at $d = 10$; the $\sim 7\%$ base-grid calibration is common to the scan and cancels from the exponent and the trend (Sec. VC). *Right, separation fixed:* (c) the equal-mass ladder at $d_0 = 20$ —four pairs whose *two* stars are retuned together, spanning a factor 2.46 in mass—returns $a \propto \bar{M}^{0.97 \pm 0.06}$ against 1 exact. (d) The same ladder compensated: each rung recovers its own mass to within 2.4%, and the residual tilt carries the finite-size sign, the lightest pair being the most diffuse. Error bars are the half-spread of four disjoint-window fits. The ladder is fitted on the halo-free core tracker; the barycentre alternative is quantified in Sec. VB.

five accelerations gives $a \propto d^{-2.028 \pm 0.011}$ against -2 exact—and adding the widest point moved the exponent *toward* the exact value (four-point fit: -2.041). The normalization is the stars’ own mass: across the scan, $a d^2/\bar{M} = 1.031 \rightarrow 1.011 \rightarrow 1.003 \rightarrow 1.001 \rightarrow 1.002$. The close-pair excess is physical and behaves like the finite-size correction it must be—largest exactly where the two stars’ envelopes overlap most, decaying monotonically, gone by $d = 16$. The $d = 20$ cell rules out a measurement floor— $a d^2$ does not plateau—and the signed residual momentum halves with each step outward too (5.8, 3.7, 2.3, 0.98, 0.49 in units of 10^{-5} at $t = 200$), tracking the acceleration rather than a diagnostic floor.

One systematic spans the whole scan: all five points share the base grid, whose acceleration at $d = 10$ sits $\sim 7\%$ below the converged fine-grid value (next subsection). The calibration is common mode and cancels from

the exponent, from the trend in $a d^2$ and from every ratio in Sec. VF, but the *absolute* accelerations—2.307, 1.448, 0.998, 0.560, 0.359 in units of 10^{-4} —carry a grid uncertainty of that size: the converged $d = 10$ acceleration, $(1.55 \pm 0.01) \times 10^{-4}$, lands 8% above $\bar{M}/d^2$, so at a gap of two stellar diameters the point-mass formula holds to ten percent rather than one. A converged separation scan is the natural sequel.

The law’s other axis is the mass. Three further pairs were built at $d_0 = 20$ —far enough out that the excess of Fig. 5b has decayed to 0.2%—with *both* stars retuned together, so each pair stays mass-matched and therefore rigid; an unequal pair deforms (Sec. VF), and a quadratic fit to a moving separation measures nothing. Equal masses need *different* frequencies in the two sectors (Fig. 2), so each phantom partner was root-found on its branch. With the archived $d_0 = 20$ cell as the fourth rung the ladder spans a factor 2.46 and returns $a \propto \bar{M}^{0.966 \pm 0.061}$, 0.6σ from unity, each rung’s own $a d^2/\bar{M}$ landing within 2.4% of 1 (mean 0.999) with the residual tilt carrying the finite-size sign (Fig. 5c,d). The exponent is quoted on the halo-free core tracker; on whole-domain barycentres it returns 1.17 ± 0.08 —the lightest, most diffuse rung sheds the most halo and its barycentre fit sits 18% low—so the two trackers bracket the exact law. Both axes of $a = \bar{M}/d^2$ are therefore direct measurements—and this one is compared rung by rung against its own prediction, not against a neighbour.

C. Convergence, and what cannot move the answer

Refining the grid does not erode the runaway—it settles it. The drift at $t = 200$ runs +2.881, +3.014, +3.002 down the ladder: the two fine rungs agree to 0.4% on drift and 0.5% on acceleration while the base rung sits 4% low, which is the opposite of what a discretization artefact does. **We quote $\Delta X(200) = 3.00 \pm 0.01$ and $a = (1.55 \pm 0.01) \times 10^{-4}$** , the fine-pair spread as the error bar. No formal convergence order accompanies this, for a measured reason (Fig. 6b): the $t = 0$ Hamiltonian norm *rises* with resolution ($2.0 \rightarrow 3.7 \rightarrow 5.6 \times 10^{-4}$, the transfer residual of Appendix A) while the evolution error falls ($1.0 \times 10^{-5} \rightarrow 5.4 \times 10^{-6}$ late-time), so the triple is non-monotone and Richardson extrapolation would return corrections smaller than the fine-pair spread at any assumed order.

Twins of the base rung close the remaining loopholes. First, the base rung is not solve-limited: a cell identical except for the elliptic tolerance—driven from $8.6 \times 10^{-4}\%$ to $1.9 \times 10^{-4}\%$, a $4.4\times$ deeper residual—changes the drift by 0.0015% and the acceleration by less than a part in 10^4 , while refining the grid moves it by 4%; the rung offset is a property of the evolution grid, not of the solve. Second, the uniform-grid choice does not shape the result: the same cell with mesh refinement enabled reproduces drift and acceleration to 0.001%–0.002%, and level 1 was never created, the tagger’s threshold ($|\chi - 1| = 0.02$) sit-

ting four times above anything these spacetimes produce. Third, neither does the gauge: doubling the Gamma-driver damping ($\eta = 1 \rightarrow 2$), one parameter and nothing else, shrinks the shift by 5.0% everywhere while the drift moves by 0.007% and the acceleration by 0.019%—the coordinates changed by three hundred times more than the physics did. Finally, doubling the box (eight times the volume, sponge moved out) changes the drift by -4.2% : the boundary is not driving the motion.

D. The mirror: swap the sectors, the runaway reverses exactly

Swapping the two sectors in place must reverse the physics and nothing else. **It reverses to two parts in 10^5** : the mirrored cell returns drift ratio -1.000022 and acceleration ratio -1.000010 against the headline cell. What that excludes is precise: any asymmetry of the machinery that is *not* sourced by the matter—a biased stencil, a lopsided box, a directional diagnostic—since none of those knows which star carries the minus sign. It does not exclude a matter-sourced artefact, which reverses under the swap just as the physics does; the retired displaced-initial-data artefact of Sec. IIIB would have passed this test. Those are excluded instead by the force law, the mass ladder and the nulls that follow.

E. The nulls, and the two interaction channels

The lone-star rows calibrate the floor: a single star, alone in the box, moves at most 1.8×10^{-3} on any axis in $t = 200$ (canonical, box centre) and 1.6×10^{-3} (phantom, released *off-centre* at $x = 37$, where no symmetry protects it)—about $2000\times$ below the signal, with no preferred direction. The pair drift is measured on barycentres, so the like-for-like floor is the phantom’s own barycentre: 1.3×10^{-2} over the same window, a factor 225 below the signal (cores alone would suggest 1600–2000); both are correct for what they track, and the null rests on the smaller ratio.

The phantom was also carried alone to $t = 1000$, five times the standard window. It reproduces the $t = 200$ control digit for digit through their overlap, then keeps going without collapsing or dispersing—but it does not hold its amplitude indefinitely: the field peak decays *linearly* at 0.632% per 100 time units from $t \approx 200$ (disjoint sub-windows 0.604, 0.648, 0.608), reaching -5.4% at $t = 1000$. Everything else stays benign: $\min \chi$ rises $1.00000 \rightarrow 1.00033$ (negative ADM mass holds χ above unity everywhere, so even the domain minimum is a hill), the rms radius grows $4.41 \rightarrow 4.83$, the confined fraction falls $0.736 \rightarrow 0.670$, and the Hamiltonian norm falls $44\times$ as the initial transient radiates. The star relaxes outward and leaks slowly; it does not go unstable. The same curve reads -0.5% at $t = 200$ and -1.6% at $t = 400$, which is why no shorter run resolves it—and why the claim here

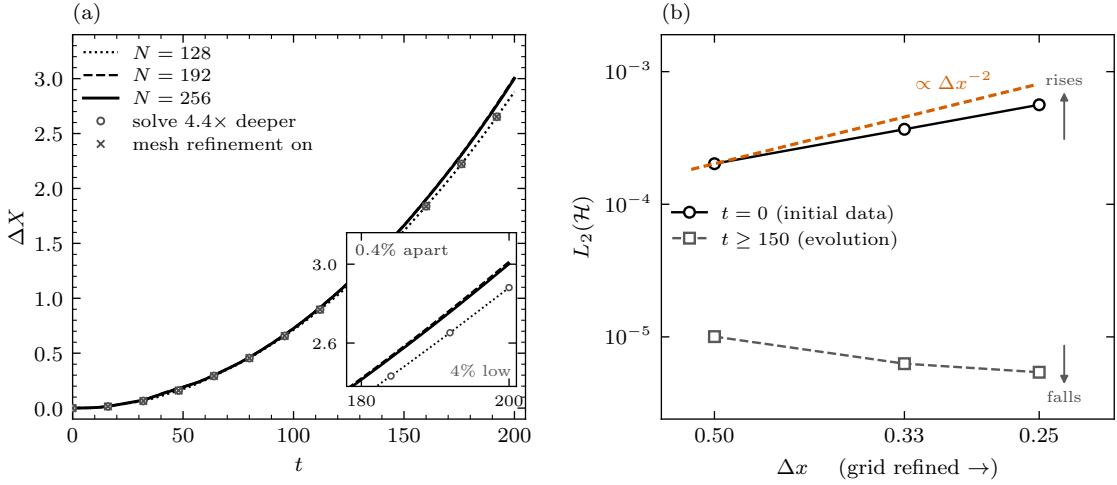


FIG. 6. (a) The resolution ladder at $d_0 = 10$: midpoint drift at $\Delta x = 0.50, 0.33, 0.25$, with two twins of the base rung overlaid—the elliptic solve driven 4.4× deeper (circles) and mesh refinement enabled (crosses); both reproduce it to 2×10^{-5} . Inset: the endpoint—the two fine grids agree to 0.4% while the base rung sits 4% low. (b) Two competing error sources: the $t = 0$ violation *rises* with resolution (the solve-to-evolution transfer residual, amplified as Δx^{-2} ; Appendix A) while the evolution’s own *falls*—two opposite grid behaviours never leave a single asymptotic power.

is a measured decay rate over the window evolved, not an asymptotic one.

The same-sign pairs are the campaign’s sharpest control, and their behaviour was not anticipated: **they merge—both of them, on the same clock**. Tracked frame by frame (Fig. 7a), the two canonical stars of PP coalesce at $t = 33.6$ and the two phantoms of MM at $t = 32.8$ —a 2.4% difference, within the trackers’ grid disparity. That timing is the measurement. Newtonian gravity *attracts* the PP pair and *repels* the MM pair—Bondi’s own sign rules, Fig. 1—so if gravity set the timescale, PP would coalesce while MM flew apart. Instead both collapse identically: the driving force is blind to the sign of the mass, and its scale gives it away—Newtonian free fall from $d = 10$ at these masses takes $t \approx 207$, so the observed $t \approx 33$ needs a force $\sim 35\text{--}40\times$ gravity. That is the same-field overlap attraction of Sec. II, present exactly and only where two lumps share one complex field; Bondi’s — repulsion is real but invisible beneath it. The pairs are seeded in phase, the attracting configuration (Sec. III B), which is what allows them to fuse into a single centred lump.

This measurement is what certifies the mixed pair as gravitational. Its two stars are *different* fields with no coupling term, so the overlap channel is structurally absent, and the data confirm that absence three ways. At the same $d = 10$ where same-field pairs close in $t \approx 33$, the mixed gap holds to 1% for $t = 200$: a few-percent leak of a $35\times$ -gravity force across sectors would have closed it visibly. The mixed force is a clean $d^{-2.03}$ power law across $d = 8\text{--}20$, where a field-overlap force dies exponentially with distance. And its strength tracks the partner’s *ADM mass* (next subsection), a quantity a field force knows nothing about.

Meanwhile the same-sign cells deliver their null un-

der the most violent history in the campaign: through plunge, merger, sevenfold ejecta growth and ringdown, their pair centroids never move more than 7.8×10^{-4} (PP) and 5.3×10^{-4} (MM) on any of six cells across three grids (Fig. 7b)—and the residual does not grow with resolution. Neither pair has a mass dipole, so neither centroid moves, *even during a merger*. The merger dynamics themselves are converged physics (ejecta growth $\times 7.1/7.0/6.9$ across the ladder), and the remnants carry the model’s structural sign difference: the PP remnant is a well in χ (min $\chi = 0.979$), the MM remnant a *hill* (χ up to 1.011)—same matter behaviour, opposite curvature.

F. Gravity scales with the source, and the gap’s sign flips at equal mass

Each star is accelerated by the *other’s* mass, so retuning one star must change its partner’s acceleration and leave its own alone. Three one-knob variants test this (Fig. 8), retuning Φ_- to $\omega = 0.88$ and 0.804 and, for the phantom-heavier rung, the *canonical* star to $\omega = 0.81$. The constant C in $\ddot{x} = \pm C/d(t)^2$ is fitted per star by regressing its core’s $x(t)$ on $\{1, t, I_2(t)\}$ over $t \geq 5$, where I_2 is the double time integral of the *measured* $d(t)^{-2}$; the fit does not assume constant separation. Only ratios of C are quoted, so the grid calibration of Sec. VB cancels. The absolute counterpart is the equal-mass ladder. Mass factors 0.600, 0.803, 0.747 return retuned-partner pull ratios 0.618, 0.810, 0.756—within 0.9–3.1% across a factor 2.2—with unchanged-partner controls 1.028, 1.011, 1.001. No rung exists below 0.60 of the matched mass: both branches floor at $|M|_{\min} \simeq 0.54$ of it (Fig. 2), which is itself a result. And the ladder carries a signature with no artefact analogue: **the gap’s behaviour flips sign ex-**

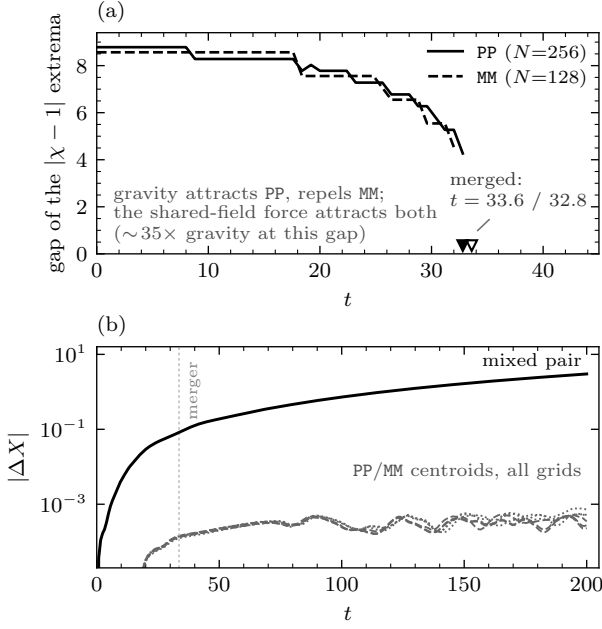


FIG. 7. The same-sign controls. (a) Gap between the two $|\chi - 1|$ extrema—wells for PP, hills for MM—tracked frame by frame: *both* pairs fall together and merge, at $t = 33.6$ and 32.8 . Newtonian gravity’s sign is *opposite* between these two cells, so their common clock is set by the sign-blind shared-field attraction, $\sim 35\times$ gravity at this gap—a channel the mixed pair structurally lacks. (b) Through plunge, merger and ringdown, the PP/MM pair centroids (all grids) stay below 8×10^{-4} —four orders under the mixed pair’s drift. Movies: Videos 3 youtu.be/jsp-Tv-f68Y and 4 youtu.be/_J8jfkUk-m4.

actly where the masses cross. Heavier phantom: the gap opens (+0.60). Matched: it holds (+0.003). Lighter phantom: it closes ($-0.59, -1.38$), monotonically. Interpolating between the two cells that bracket the flip puts it at mass ratio 0.995 ± 0.03 , against 1.000 predicted; the error bar is the spread over end-of-run gap conventions (barycentre 0.995, proper 0.994, core 0.961, late-window mean 1.002), which dominates the statistical one. No grid, boundary or solver distinguishes which of the two stars is heavier; gravity does.

VI. LIMITS ON GRAVITATIONAL RADIATION AND NEAR-ZONE BEHAVIOUR

A uniquely relativistic aspect of the runaway is its radiation: the exact accelerating-pair metrics are radiative [2, 3], and negative-mass binaries have been proposed as sources of anomalous signatures [39], with the leading anomalous channel for mismatched charge-to-inertia binaries being dipolar. The bicomplex model closes that channel structurally: the source of gravity is the signed stress tensor, conserved on shell (Sec. II), so the rate of change of the signed mass dipole is the conserved signed momentum—zero for this data, and measured to cancel

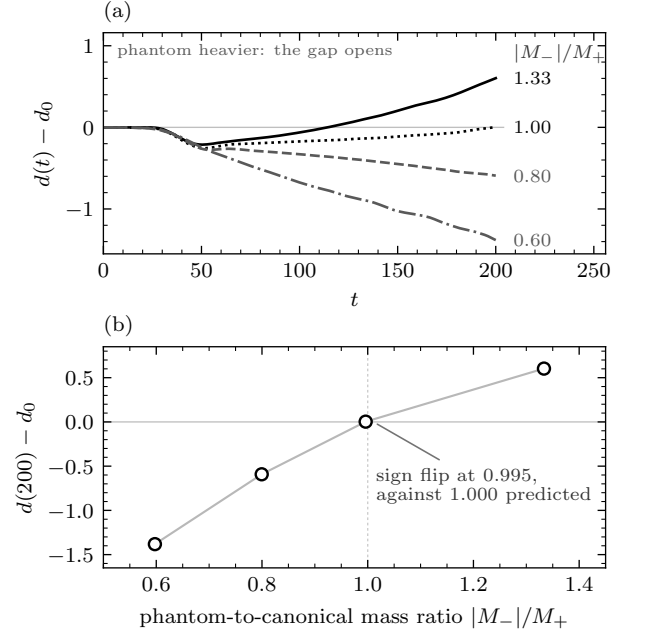


FIG. 8. The mass ladder: the matched pair and three retuned partners spanning a factor 2.2 in mass ratio. (a) Gap history: with a heavier phantom the gap *opens*, matched it holds, with a lighter phantom it *closes*, monotonically in the ratio. Every cell shares the same $t \approx 50$ gauge dip, so the gap is read late. (b) Gap change at $t = 200$ against the pair mass ratio $|M_-|/M_+ = 0.597, 0.800, 0.996, 1.333$, monotone through the four cells: the sign flip falls at ratio 0.995, against 1.000 predicted.

to $\lesssim 1\%$ (Sec. V A)—and mass-dipole radiation is forbidden at leading order for exactly the reason it is forbidden in ordinary general relativity. Kinematics agrees from the other side: Ψ_4 carries spin weight -2 , so its expansion begins at $\ell = 2$ whatever the dipole does. What survives is quadrupolar bremsstrahlung from the accelerating pair—secular, chirpless, and for this configuration very small. The signed mass quadrupole of a zero-momentum pair is $Q \simeq 2\bar{M}dX(t)$, so at constant separation and constant acceleration $\ddot{Q} = 2\bar{M}ad$ is *constant*: measured at 4.5×10^{-5} against 4.4×10^{-5} predicted, steady to 7% over $20 \leq t \leq 190$. A constant $\ddot{Q}$ radiates nothing at leading order, and the flux enters only through the slow secular evolution of d and v —from the trajectories $|\dot{Q}| \leq 6 \times 10^{-8}$, a quadrupole luminosity $\sim 4 \times 10^{-17}$ and $\sim 7 \times 10^{-15}$ radiated over the run, ten orders of magnitude below the pair’s total mass of 5.5×10^{-5} . So the $\ell = 2$ content is bounded rather than measured here, every extraction shell the domain affords lying inside the near zone.

The doubled box measures how much of that reaches the wave zone at these speeds: **none that we can detect out to $R = 40$** (Fig. 9). The $\ell = 2$ amplitude falls as $r^{-4.8}$ across shells spanning $R = 16\text{--}40$ — $r\psi_4$ drops by a factor 31 where one outgoing wave would keep it flat—so every shell sits in the near zone, reading the

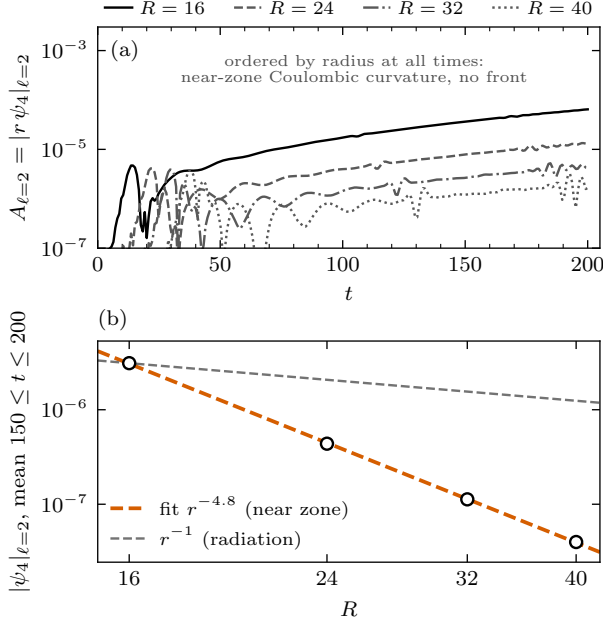


FIG. 9. The wave-zone measurement on the doubled box ($L = 128$, $\Delta x = 0.25$, sponge at 48–60). (a) Total $\ell = 2$ amplitude $A_{\ell=2}$ on the four extraction shells: ordered by radius at all times, with no propagating front. (b) The falloff, averaged over $150 \leq t \leq 200$: $|\psi_4|_{\ell=2} \propto r^{-4.8}$, against r^{-1} for radiation. Every shell out to $R = 40$ is in the near zone, so $A_{\ell=2}$ is a Coulombic near-field amplitude, not a radiated flux—consistent with the quadrupole estimate of Sec. VI, ten orders of magnitude below this bound.

source’s Coulombic curvature rather than a flux. This is a null with a number, and it agrees with the structural argument above; reaching the radiative regime needs relativistic speed—only $0.03c$ inside the fit window here (Sec. VIII).

VII. SYSTEMATICS, WITH THEIR MEASURED SIZES

How accelerations are measured. Every quoted acceleration is a parabola fit to the midpoint trajectory from $t = 5$ (Appendix A) to the end of the run. Fitting only the last two thirds of a run instead changes any single a by 1–3%; the trends—the d^{-2} exponent, the mass-law ratios, the sign flip at equal mass—do not change.

Grid calibration. The base grid underestimates the acceleration by $\sim 7\%$ at $d_0 = 10$; the two fine grids agree to 0.5% and set the quoted values. The base-grid scans all carry the same offset, which cancels from their ratios and exponents (Sec. VB).

The gap opens after $t = 200$, and why. Over $t \leq 200$ the separation holds to 1% (the two trackers bracket its change between -0.9% and $+0.1\%$). On the base grid it then opens: $d = 11.8$ at $t = 400$, 18.2 at $t = 600$, 29.8 at $t = 784$, growing roughly as $(t - t_0)^{2.8}$ with $t_0 \simeq 110$.

What opens it is one star. The trailing phantom’s acceleration falls steadily, reaches zero near $t \approx 530$ and reverses, standing at -7.5×10^{-5} by $t = 750$ while the leading star still accelerates at $+1.1 \times 10^{-4}$. **After $t \approx 500$ this is one runaway star with a slowing companion, not a pair**, and no pair quantity is quoted beyond that point. (The common motion still obeys the force law: $a d^2$ stays within 35% of $\bar{M}$ over the whole run while a falls nine-fold.)

The separation is fragile for a structural reason. With matched masses both stars accelerate equally at *every* separation ($\ddot{x}_+ = |M_-|/d^2$, $\ddot{x}_- = M_+/d^2$), so nothing pulls d back toward d_0 : any small difference between the two accelerations, from any source, grows without limit, while the midpoint drift that measures the runaway is unaffected. Three sources were tested. *Unequal masses*—no. Three of the four equal-mass pairs are matched to better than 0.02% and still open by $+0.009$ to $+0.030$ at $t = 200$; the headline pair’s own 0.39% mismatch has the wrong sign (it would close the gap by 0.011; the converged rung opens by $+0.013$). *The moving box*—no. A recentring box run to $t = 784$ and a static box started ten units off centre agree on the separation to 0.5% through $t = 500$ and on their field contents to 0.1%; the comparison stops at $t = 500$ because the static box’s leading star reaches the sponge at $t = 600$. *The grid*—yes. The opening at $t = 200$ falls from $+0.107$ to $+0.029$ to $+0.013$ as Δx goes from 0.50 to 0.25, an 8.4-fold drop for a factor two in spacing: it converges away, as a physical separation change would not. At fixed grid it scales as $\bar{M}^{1.3}$ and $d^{-2.9}$, like a field gradient across the star (M/d^3) rather than the force (M/d^2). A plausible seed is that the discretized stars are not exactly stationary—their field content grows by 21.0, 16.1 and 12.4% by $t = 200$ down the ladder, and by the same 20.0 and 20.9% for the lone stars—and the two sectors need not drift identically; we record this as a candidate, not a demonstrated mechanism.

What follows is limited. The $t \leq 200$ results are unaffected: they are quoted at the converged rung, where the opening is 0.13%. Beyond $t = 200$ the base-grid separation is unconverged, and the grid-error attribution is an extrapolation—from an opening of 0.107, where the ladder was measured, to 8.2 at $t = 600$, a factor 77—so no quantitative conclusion is drawn from it. One late-time run at $N = 192$ would decide it; the prediction it must meet is $d(400) \simeq 10.7$, against 11.85 on the base grid. The speeds reached in the long cells—the leading star at $0.10c$ near $t = 600$ and $0.12c$ at $t = 784$, the midpoint at $0.081c$ —come only after rigidity is gone. Relativistic speed needs a finer grid, not a longer run.

Gauge. Drift and gap are coordinate quantities, so the gauge is checked directly. The shift at the two cores never exceeds 7.7×10^{-4} and is nearly antisymmetric (β^x points one way at the canonical star and the other at the phantom), so its mean—the only part that could move the pair as a whole—stays below 1.1×10^{-5} against a midpoint velocity of 3.0×10^{-2} ; integrated over the run

it accounts for 4.0×10^{-4} of displacement, 0.013% of $\Delta X = 3.00$. The drift is matter moving through the Eulerian frame, anchored at the Sommerfeld boundary, not coordinates being dragged; the $\eta = 2$ twin of Sec. VC confirms it by run. Nulls are quoted on cores because a whole-domain barycentre picks up domain noise (the off-centre lone phantom’s wobbles by 4×10^{-2} early while its core never moves more than 1.6×10^{-3}); the pair is quoted on both trackers, which agree to 1%.

Same-sign initial data. The conformally flat outer boundary condition of the elliptic solve is wrong for a box carrying net mass $2M$, so the same-sign solves floor near $10^{-3}\%$ instead of converging further. This is a $t = 0$ statement only—their evolution constraints stay bounded through the merger—and is why those cells serve as momentum nulls, not precision measurements.

VIII. DISCUSSION AND CONCLUSION

We posed Bondi’s positive–negative mass pair as a full initial-value problem, and it behaved as he argued in 1957—quantitatively.

(1) The runaway is real, and it is steady. Released at rest, the mixed pair self-accelerates at a rate flat to 2% through $t = 400$, reaching 0.056c; at the base grid the rate eventually falls as discretization error spreads the pair, and $a d^2$ goes on tracking $\bar{M}$ in the pair’s common mode, though not in the differential mode that opens the gap; rigidity is not quoted past $t \approx 500$ (Sec. VII). Swapping the sectors reverses the drift to two parts in 10^5 ; the separation holds to 1% over $t \leq 200$, in proper distance as in coordinates.

(2) It is gravity. By construction the sectors meet only in the metric; by measurement the force falls as $d^{-2.03 \pm 0.01}$ with $a d^2$ returning the stars’ mass, and on a mass-matched ladder spanning a factor 2.46 the mass exponent itself comes back as 0.97 ± 0.06 ; differentially, the pull tracks the partner’s ADM mass through a factor 2.2—including the gap’s sign flip at equal mass. The same-sign controls isolate the model’s only other channel, the shared-field overlap attraction, and show it $\sim 35\times$ gravity, short-range and sign-blind—present where the model puts it, absent from the mixed pair.

(3) Displacement without momentum—the Bondi signature. The sector momenta, each growing to 5×10^{-4} , cancel in the signed sum to under 1%, the residual converging toward zero with resolution while the displacement converges onto 3.00 ± 0.01 . Same-sign pairs, having no mass dipole, hold their centroids at the 10^{-4} level *even while merging*.

(4) It does not detectably radiate, as predicted. The signed dipole cannot radiate—its rate of change is the conserved signed momentum—and the constant- $\dot{Q}$ quadrupole radiates nothing at leading order: an estimated 7×10^{-15} over the run, with the measured $\ell = 2$ amplitude falling as $r^{-4.8}$ out to $R = 40$ —near field, not flux (Sec. VI).

(5) A star of negative ADM mass exists as constraint-satisfying, asymptotically flat initial data, and survives evolution—400 units paired, 1000 alone, its peak flat to 0.5% over the standard $t \leq 200$ window (static phantom-field stars [31] and de Sitter bubbles [29] were known; a negative-mass *horizon* is a different object [12]). It is the better-behaved of the pair, shed field being pushed away rather than re-accreted. Alone to $t = 1000$ it neither collapses nor disperses but relaxes outward, its central amplitude falling linearly at 0.63% per 100 time units: a measured decay rate over the window evolved, not asymptotic stability.

What is not established. Four things remain open. (i) The absolute strength of the force away from $d = 10$: the separation scan ran only at the base grid, which reads $\sim 7\%$ low (Sec. VII), so the exponent is secure but the normalization elsewhere is not. (ii) Rigidity past $t = 200$: on the base grid the trailing star slows and the gap opens. Every test in Sec. VII points to a grid error that shrinks with resolution, acting on a separation that nothing restores—but that is extrapolated a factor 77 beyond where the resolution ladder was measured, and a late-time run at $N = 192$ would decide it. (iii) Stability against sideways perturbations: every release here is head-on. (iv) The radiation. The near-zone null agrees with the no-dipole argument and the quadrupole estimate of Sec. VI, but the question this configuration was built to reach—what happens to the energy of a radiating system whose total mass is only 5.5×10^{-5} , 0.4% of either star, with no positive-mass theorem to stop it falling—needs relativistic speed, and speed is now limited by resolution, not run time: $\sim 0.3c$, where the anomalous-signature programme of Ref. [39] meets a concrete source, is a converged-grid measurement waiting to be made. Nothing here bears on whether such matter exists.

IX. REPRODUCIBILITY

Every number and figure derives from the packed campaign under `results/bondi-dipole-runaway` of the repository [44]: one directory per cell under `campaign/`, each holding the tracker streams, constraint norms, Weyl modes, solver history, and the exact launch environment. Cell names carry the configuration (`runaway_pair_d10_L64_N256_lev0`: the mixed pair at $d_0 = 10$, $L = 64$, 256^3 , uniform), and `stars/` holds the dressed-star profile tables and both family scans. A single script (`article_figures.py`, in the repository’s visualisation package) regenerates every figure and prints every quoted number. The code lives in the same repository [44], tagged at the version behind this paper; an archived snapshot with a DOI accompanies the published version. See Supplemental Material at [URL will be inserted by publisher] for movies of the headline cell (Video 1), its $t = 400$ continuation (Video 2) and both same-sign controls (Videos 3 PP and 4 MM). Initial data:

GR_{Tresna} [25]; evolution: GR_{Teclyn} [23, 24, 26].

ACKNOWLEDGMENTS

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Appendix A: Constraint behaviour

Initial data. The GR_{Tresna} [25] solves themselves are clean: every mixed-pair solve converges monotonically on the maximal-slicing path and exits under its tolerance gate, with residuals of $8.6 \times 10^{-4}\%$ (Hamiltonian) and $8.0 \times 10^{-4}\%$ (momentum) at the base grid after 13 nonlinear iterations, tightening to $8.7 \times 10^{-5}\%$ and $5.1 \times 10^{-5}\%$ on the finest rung (16 iterations). The gate scales with the grid, so a fine rung never inherits a coarse rung’s residual, as it would on a fixed-tolerance ladder.

The solve is not, however, what sets the $t = 0$ violation on the *evolution* grid: the deep-solve twin of Sec. V C drove the solve $4.4\times$ deeper at fixed grid and lowered that violation by only $\sim 1\%$ ($2.02 \rightarrow 2.00 \times 10^{-4}$). What dominates is the solve-to-evolution transfer, set by the hand-off order of Sec. III B: the handoff leaves a metric–matter mismatch at the grid scale, whose amplitude does not fall with resolution, and the constraint operator reading it is a second difference carrying Δx^{-2} . No solver tolerance touches that term; removing it would mean matching the two discretizations at the order of the constraint operator itself.

The transfer error is bounded rather than removed, and transient: a $t = 0$ spike that radiates away, falling by an order of magnitude within $t \lesssim 3$ —faster on finer grids, as a grid-scale transient should—and settling onto its late plateau by $t \approx 4$, before the $t \geq 5$ fit window opens.

Evolution. Figure 10 shows the headline ladder in absolute code units (the domain is overwhelmingly vacuum, so relative norms are dominated by near-zero denominators). Each rung relaxes from its initial-data spike within $t \lesssim 2$ and then holds nearly flat for the whole run: late-time $L_2(\mathcal{H})$ of 1.0×10^{-5} , 6.3×10^{-6} , 5.4×10^{-6} down the ladder—falling with resolution while the $t = 0$ spike rises, the two-error-source structure of Fig. 6b. The zero initial

shift also emits a flat $\sim 10^{-6}$ gauge pulse inward at $t = 0$ (visible in the packed shift frames [44]); it crosses the stars by $t \approx 20$ and leaves no trace in the constraints or the fitted acceleration. The PP control runs a factor few above the mixed cell after its merger and stays bounded to $t = 200$, its centroid null holding throughout.

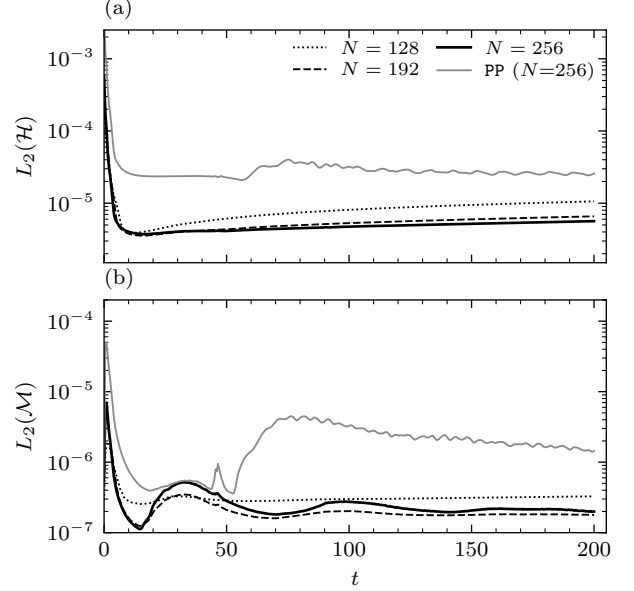


FIG. 10. Constraint norms for the headline cell’s resolution ladder (line styles as in Fig. 6) and the merging PP control at the finest grid (gray). (a) Hamiltonian, L_2 over the domain: finer grids start worse (initial-data interpolation noise) and evolve better. (b) Momentum, L_2 .

Appendix B: Computational cost

Every evolution ran on one NVIDIA H100 80 GB device of a four-H100 node with two 32-core Xeon Platinum 8462Y+ processors; the elliptic solves ran CPU-side on 32 MPI ranks, ~ 20 minutes at 256^3 and ~ 4 hours at 512^3 . GR_{Teclyn} was built against AMReX with CUDA 12.1 and GCC 11.4; each cell’s metadata records its build commit. At $\Delta t = 0.02 \Delta x$, a $t = 200$ evolution takes 20 000/30 000/40 000 steps at $N = 128/192/256$, using $\sim 6/20/48$ GB of device memory (uniform 256^3 fits on one card) and 1.1, 4.8–6.0 and 14.6 hours of wall time, the last including the movie/still pipeline. The wave-zone box ($L = 128, 256^3$) takes 7.4 h, the $t = 400$ long run with frames 3.8 h, the lone star to $t = 1000$ 5.4 h, and the moving-box pair to $t = 784$ 4.3 h. The thirty-four packed cells total $\simeq 99$ GPU-hours of evolution.

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